

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

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Aim: Tree Traversal

Q. Implement pre-order, Post-order and In-order traversal in a binary tree

Code:

```
class Node:
```

```
    def __init__(self, item):
```

```
        self.left = None
```

```
        self.right = None
```

```
        self.val = item
```

```
def inorder(root):
```

```
    if root:
```

```
        inorder(root.left)
```

```
        print(str(root.val) + "->", end="")
```

```
def postorder(root):  
    if root:  
        postorder(root.left)  
        postorder(root.right)  
        print(str(root.val) + "->", end="")
```

```
def preorder(root):  
    if root:  
        preorder(root.left)  
        preorder(root.right)  
        print(str(root.val) + "->", end="")
```

```
root = Node(1)  
root.left = Node(2)  
root.right = Node(3)  
root.left.left = Node(4)  
root.left.right = Node(5)
```

```
print("Inorder traversal")  
inorder(root)  
print("\npostorder traversal")  
postorder(root)  
print("\npreorder traversal")
```

preorder(root)

Output:

```
DS/prac6.py
Inorder traversal
4->2->1->
postorder traversal
4->5->2->3->1->
preorder traversal
4->5->2->3->1->
```